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August 16, 1966
SUPERSEDING
Fed. Spec. ZZ-H-141b
August 9, 1961

FEDERAL SPECIFICATION

HEELS, RUBBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heels made of natural rubber, synthetic rubber, or a mixture of these materials.

1.2 Classification.

1.2.1 *Types, classes, and styles.* The rubber heels shall be of the following types, classes, and styles, as specified (see 6.2).

Type I—Half heels.
Class:
1—Standard composition.
3—Nonmarking composition.
5—Wood core heel, oil resistant, nonmarking composition.
6—Wood core heel, nonmarking composition.
7—Plastic core heel, oil resistant, nonmarking composition.
8—Plastic core, nonmarking composition.
Style 1—Straight breast.

Type II—Whole heels.
Class:
1—Standard composition.
2—Oil resistant composition.
3—Nonmarking composition.
4—Oil resistant, nonmarking, cork composition.
5—Wood core heel, oil resistant, nonmarking composition.
6—Wood core heel, nonmarking composition.
7—Plastic core heel, oil resistant, nonmarking composition.
8—Plastic core heel, nonmarking composition.
Style:
1—Straight breast.
2—Full beveled breast.
3—Straight breast truncated corner.

Type III—Junior and cuban heels.
Class:
1—Standard composition.
3—Nonmarking composition.
Style 1—Straight breast.

1.2.1.1 *Subclass (a).* When specified (see 6.2), types I and II heels in classes 3, 5, 6, 7, and 8 shall be supplied in high abrasive composition (subclass (a)).

1.2.2 *Sizes.*

1.2.2.1 *Types I and II.* Heels furnished to shoe manufacturers for use on new shoes and boots shall be sized in B&S (USMC) sizes. Type I heels shall be furnished in sizes 24 through 42, as specified. Type II heels shall be furnished in sizes 27 through 41, as specified. Heels furnished to Government agencies for repair use shall be dealer sized. Type I shall be furnished in sizes 7-8 through 13-14, as specified. Type II shall be furnished in sizes 7-8 through 14-15, as specified (see 6.2).

1.2.2.2 *Type III.* Type III cuban heels shall be furnished in sizes 1, 3, 5 and 7, as specified; type III junior heels shall be furnished in sizes J-14 through J-17, as specified (see 6.2).

1.2.3 *Colors.* Rubber heels shall be of the following colors, as specified (see 6.2):
Color No. 1—Black.
Color No. 2—Tan.

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Color No. 3—Neutral.
Color No. 4—White.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-T-45—Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing.
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 601—Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-B-43291—Boxes: Fiberboard; Corrugated, Double-Wall Weather Resistant.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

Drawings:

Army Natick Laboratories
8-1-160—Heels, Rubber; Type II, style 2 (Full Beveled Breast).

(Miniature copy of Drawing 8-1-160, identified as figure 1, is attached for information purposes only.)

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D-1484-59—Penetration of Hard Rubber by Type D Durometer.
D-1630-61—Tentative Method of Test for Abrasion Resistance of Rubber Soles and Heels.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race St., Philadelphia, Pa., 19103.)

U.S. Department of Commerce Commercial Standard:

CS35-61—Hardwood, Plywood.

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, one pair of each type, class, style, size, and color of heels being produced, shall be submitted to the contracting officer or his authorized representative for inspection as specified in 4.2. The approval of the pre-

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production sample authorizes the commencement of production but does not relieve the supplier of the responsibility for compliance with all provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production items.

3.2 Material.

3.2.1 *Classes* 1, 2, 3, 5, 6, 7, *and* 8. Classes 1, 2, 3, 5, 6, 7, and 8 shall be made of natural rubber, synthetic rubber, or a mixture of natural and synthetic rubber. The type rubber used, or the proportions of each type rubber if mixtures are employed, shall be such as to meet the requirements specified for each class, as applicable (see 1.2 and 3.3).

3.2.2 *Class* 4. Class 4 heels shall be made from a mixture of synthetic rubber and dry cork of 14/24 grade. The mixture shall consist of 8 ± 2 percent by weight of cork, the balance shall be rubber compound.

3.3 Physical characteristics.

3.3.1 *Abrasive index before and after aging.*

3.3.1.1 *Classes* 1, 2, 3, *and* 4. The abrasive index before aging shall be not less than 50 for classes 1, 2, and 3 and not less than 35 for class 4 heels, when tested as specified in 4.3.3. After aging as specified in 4.3.3, the abrasive index of classes 1, 2, 3, and 4 heels shall be not less than 75 percent of the abrasive index determined on the heels before aging.

3.3.1.2 *Classes* 5, 6, 7, *and* 8. The abrasive index for classes 5, 6, 7, and 8 shall be not less than 65 before and after aging when tested as specified in 4.3.3.

3.3.1.3 *Subclass (a) heels.* The abrasive index for subclass (a) heels shall be not less than 175 before and after aging when tested as specified in 4.3.3.

3.3.2 *Hardness (before and after aging).* The hardness of the heels, (except classes 5, 6, 7, and 8) shall be not less than 60 nor more than 80. The hardness of classes 5, 6, 7, and 8 shall be not less than 50 nor more than 75 when tested as specified in 4.3.3. After aging as specified in 4.3.3, the hardness of classes 1, 2, 3, and 4 heels shall be not more than 10 points higher than the hardness determined on the heels before aging.

3.3.3 *Volume swell (classes* 2, 4, 5, *and* 7). Class 4 heels shall have a volume swell not greater than 65 percent, and classes 2, 5, and 7 shall have a volume swell not greater than 50 percent when tested as specified in 4.3.3.

3.3.4 *Nonmarking composition (classes* 3, 4, 5, 6, 7, *and* 8). Classes 3, 4, 5, 6, 7, and 8 shall produce only a slight easily erasable mark on white paper when tested as specified in 4.3.3. The heels shall not contain over three parts of carbon black per hundred parts of rubber.

3.4 Thickness. The thickness of the heels shall be in accordance with table I when measured as specified in 4.3.3.

TABLE I. *Thickness of heels*

Heel		Thickness (inch)
Type I	— Back center.	1/2 ± 1/64
	Center breast.	7/16 ± 1/64
Type II	— Back center.	7/8 min.
	Center breast.	5/8 min.
Type III	— Back center.	3/8 min.
	Center breast.	5/16 min.

3.5 Design. The molded heels shall be of commercial design conforming to types and styles as specified (see 1.2), except that design and dimensions of type II, style 2 heels shall conform to Drawing 8-1-160, and the dimensions of the type I heels shall be as specified in figure 2.

3.5.1 *Nail holes.* Each nail hole shall be provided with a metal washer molded in the heel.

3.5.1.1 *Type I (half heels).* Classes 1 and

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3 half heels shall have 8 holes properly located. There shall be no nail holes in classes 5, 6, 7, and 8.

3.5.1.2 *Type II, classes* 1, 2, *and* 3. Type II, classes 1, 2, and 3 whole heels shall have 13 nail holes. The 5 rear nail metal washers shall be molded in the heel 1/16 inch lower than the 8 forward nail metal washers. All nail holes shall be properly located.

3.5.1.3 *Type II, class* 4. There shall be no spaces, nail holes, nor metal washers in the heel.

3.5.1.4 *Type II, classes* 5, 6, 7, *and* 8. There shall be no nail holes or washers in classes 5, 6, 7, and 8.

3.5.1.5 *Type III (junior and cuban heels).* Unless otherwise specified, each junior heel shall have not less than 7 nail holes. Each cuban heel shall have not less than 5 nail holes. Nail holes for junior and cuban heels shall be properly located.

3.5.2 *Heel cores.*

3.5.2.1 *Types I and II, classes* 5 *and* 6. The wood core shall be a 3-ply plywood, black gum, 3/16 inch thickness. The plywood shall conform to the requirements for type I (exterior), good grade (1), of CS 35-61. The bonding agent between the plies shall be treated with a suitable bacteriacide or fungicide to inhibit degradation. The exposed face shall be coated with a suitable waterproofing substance. The distance of the core from the rear heel edge shall be 5/16 + 1/32, —1/16 inch. A rubber overlap not less than 5/16 inch shall be bonded and encompass the exterior periphery of the heel core. The core or its plies shall neither delaminate nor separate from the heel when tested as specified in 4.3.3.

3.5.2.2 *Types I and II, classes* 7 *and* 8. The plastic core shall be not more than 5/16 inch nor less than 1/8 inch from the edge of the heel, from breast corner to breast corner, except that the distance from the breast edge may be closer than 1/8 inch. The depth of the core shall be 8/32 ± 1/32 inch. The hardness shall be 65 ± 5 Shore D and not less than 55 Shore D after oil immersion. The core shall be firmly bonded to the rubber and shall show no evidence of separating before and after oil immersion, nor evidence of cracking when tested as specified in 4.3.3. The exposed surface of the core shall be stippled.

3.6 Sizes. Sizes shall be in accordance with 1.2.2 and as specified in the contract or order (see 6.2).

3.7 Color. The color shall be color No. 1, 2, 3, or 4 as specified (see 1.2.3 and 6.2).

3.8 Marking for identification.

3.8.1 *Trademark and size.* The manufacturer's name or trademark shall be legibly marked on the wearing surface of each heel. Both the B&S size and the dealers size shall be legibly marked on the attaching face of each heel.

3.9 Workmanship. The finished heels shall be clean, well made, and shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When required, the preproduction sample submitted in accordance with 3.1 shall be in-

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spected for all the provisions of this specification applicable to end product examination and tests.

4.3 Inspection. Inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated.

4.3.1 *Component and material inspection.* In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3.2 *Inspection of the end item.* The end item shall be examined in accordance with the classification of defects at the inspection levels and AQLs set forth in 4.3.2.3. The lot size shall be the number of pairs of heels of the same type, class, style, and color submitted for inspection at one time.

4.3.2.1 *Visual examination.* The heels shall be examined in accordance with the list of defects set forth below. The sample unit shall be one completely fabricated pair of heels.

Examine	Defect
Pairing	Improperly matched, i.e., not same size or wide variation in color or appearance.
Size	Not as specified.
Color	Not as specified; streaked, mottled.
Design	Any characteristic not in accordance with the specified requirements.
Type, class, and style	Not as specified.
Construction and workmanship	Any area of porosity or sponginess resulting in loss in density of the rubber stock.
	More than one cavity or air check 1/16 inch or more deep but not exceeding 1/3 the thickness dimension of the heel.
	Any cavity or air check more than 1/3 the thickness dimension of the heel in depth.
Construction and workmanship (cont'd)	Any press cut, tear, check, blister, loss of tread detail, imbedded foreign matter, or otherwise defective.
	Not cleanly trimmed.
	Less than specified number of nail holes.
Identification marking and size marking	Missing; incomplete; incorrect; illegible.
Heel cores (applicable to plywood and plastic)	Material other than specified.
	Any core omitted, damaged, cracked, delaminated, or improperly bonded.
Plywood core	Heel overlap missing or not bonded as specified.
	Waterproof coating missing or improperly applied.
Plastic core	Any plastic core showing evidence of separation or softening.
	Core not stippled.

4.3.2.2 *Dimensional examination.* The heels shall be examined for defects in dimensions in accordance with the list below. The sample unit shall be one heel.

Defect
Any dimension that does not conform to figure 2 (type I).
Any dimension that does not conform to Drawing 8-1-160, as applicable (type II, style 2).
Any wood core heel dimension that does not comply with the requirements of 3.5.2.1.
Any plastic core heel dimension that does not comply with the requirements of 3.5.2.2.

4.3.2.3 *Inspection levels and AQLs.* The inspection levels and the AQLs expressed in defects per hundred units shall be as follows:

	Inspection level	AQL
For examination in 4.3.2.1	II	4.0
For examination in 4.3.2.2	S-2	4.0

4.3.3 *Testing of the end item.* The methods of testing contained in Fed. Test Method Std. No. 601, whenever applicable and as listed in table III shall be followed. The physical values specified apply to the average of determinations made on a sample unit. All test reports shall contain the individual

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values utilized in expressing the final result. The sample unit for test shall be twelve heels. The sample size (number of sample units) shall be as shown in table II. The lot shall be unacceptable if one or more units fail to meet any requirement specified. For those characteristics where test results are reported as pass or fail, there shall be no failure of any test specimen to meet the requirement as specified. For requirements applicable to the lot average, there shall be no failure to meet the requirement as specified.

TABLE II

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. *End item tests*

Characteristic	Requirement paragraph	Test method	Requirements applicable to		Number determinations per unit	Results reported as	
			Individ. unit	Lot aver.		Pass or fail	Numerically to nearest
Composition	3.2.1 and 3.2.2	[1]	—	—	—	—	—
Nonmarking composition (classes 3, 4, 5, 6, 7, and 8):							
Marking test	3.3.4	4.4.1	X	—	1	X	—
Carbon black content	3.3.4	[1]	—	—	—	—	—
Thickness	3.4	[2]	X	—	3	—	1/64 inch
Abrasive index before aging	3.3.1	ASTM D-1630-61	—	X	3	—	Unit
Abrasive index after aging	3.3.1	7221[3] and ASTM D-1630-61	—	X	3	—	Unit
Hardness before aging	3.3.2	3021	—	X	5	—	Scale reading
Hardness after aging	3.3.2	3021 and 7221[3]	—	X	5	—	Scale reading
Volume swell (classes 2, 4, 5, and 7)	3.3.3	6211[4]	X	—	3	—	Percent
Plywood cores							
Material	3.5.2.1	[1]	—	—	—	—	—
Bond test	3.5.2.1	4.4.2	X	—	1	X	—
Plastic cores:							
Hardness, Shore D durometer							
before oil immersion	3.5.2.2	ASTM D-1484-59	—	X	5	—	Scale reading
after oil immersion	3.5.2.2	4.4.3.2	—	X	5	—	Scale reading
Cracking	3.5.2.2	4.4.3.1	X	—	1	X	—
Bond test	3.5.2.2	4.4.3.2	X	—	1	X	—

[1] The supplier shall furnish a certificate of compliance for each shipment or lot citing conformance to the applicable requirement.

[2] Cut heel in half lengthwise; use method 2021 of Fed. Test Method Std. No. 601 to measure thickness. Alternatively, a ratchet type machinist micrometer may be used without cutting heel.

[3] Aging shall be conducted at 212°F. for 24 hours.

[4] Specimens shall be cut from heel, shall be 1 by 1 inch, and shall be buffed to a thickness of 0.075 ± 0.010 inch. The immersion medium shall be medium No. 6 fuel of method 6001 of Fed. Test Method Std. No. 601.

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4.3.4 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing.
	Any component damaged affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling.
	Bulged or distorted container.
Content	Number per interior package or container is more or less than specified.

4.4 Test methods.

4.4.1 *Marking test.* The edge of the heel shall be drawn across a sheet of white bond paper using moderate pressure. If only a light mark is made which is easily erased by rubbing with the fingers, the heels shall be classified as nonmarking.

4.4.2 *Wood core bond test.* The heel shall be immersed in medium No. 6 fuel of method 6001 of Fed. Test Method Std. No. 601 for four hours at standard conditions. The heel shall then be removed from the fluid, wiped, and the overlap shall be lifted. Neither the core nor overlap shall show evidence of separation, delamination, or lack of bonding.

4.4.3 *Plastic core tests.*

4.4.3.1 *Cracking, low temperature.* The heel containing the plastic core shall be conditioned as specified in method 5111 of Fed. Test Method Std. No. 601 to an equilibrium temperature of 0°F. ± 1.8°F. The specimen shall be removed after 3 hours. Immediately thereafter, 5 heeling nails shall be immediately hammered by hand into the exposed core to a depth greater than 5/16 inch. The nails shall be placed more than 1/2 inch apart along the rear center and sides and the core shall show no evidence of cracking.

4.4.3.2 *Bond and hardness.* The stippled surface of the plastic core shall be sanded smooth and the heel shall be immersed in medium No. 6 fuel of method 6001 of Fed. Test Method Std. No. 601 for 4 hours at standard conditions. The heel shall be removed from the fluid, wiped, and examined. The core shall show no evidence of separation, delamination, or lack of bonding. The core shall then be tested for hardness on its smoothened surface in accordance with ASTM method D-1484-59, Shore D durometer.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A* (see 6.4 and 6.5).

5.1.1.1 *Types I and II.* Packaging not required.

5.1.1.2 *Type III.* One hundred and forty-four pairs of heels of one description only shall be packaged in a fiberboard interior box conforming to type CF, class domestic, variety SW, grade 175 of PPP-B-636. The approximate inside dimensions of each interior box shall be as specified in table IV. The heels, in pairs, shall be arranged on end 12 in length, 4 in width and 3 in depth within the box. Each box shall be securely closed with 2 inch minimum width gummed paper tape conforming to type III, grade A of PPP-T-45.

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TABLE IV. *Interior box dimensions for type III heels*

Heels	Sizes	Approximate inside dimensions[1] (inches)
Cuban	1	9-1/2 by 7 by 6
	3	9-1/2 by 8 by 7
	5, 7	9-1/2 by 9 by 7-1/2
Junior	J-14 through J-17	9-1/2 by 11-1/2 by 9-1/4

[1] Furnished as a guide only.

5.1.2 *Level C.* Heels shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.*

5.2.1.1 *Types I and II.* One hundred and forty-four pairs of type I heels or 72 pairs of type II heels of one description only shall be packed in a fiberboard shipping container as hereinafter specified. The approximate inside dimensions of each shipping container shall be as specified in table V. The heels, in pairs, shall be arranged on end within the shipping container as specified in 5.1.1.2, except that type II heels shall be arranged 6 pairs in length.

TABLE V. *Shipping container dimensions for types I and II heels*

Heels	Sizes	Approximate inside dimension[1] (inches)
Type I—half heels	7-8, 9-10	13-1/4 by 11-1/4 by 9-1/4
	10-11, 11-12	13-1/4 by 12-1/2 by 10-1/4
	12-13, 13-14	13-1/4 by 13-3/4 by 11-1/2
Type II—whole heels	7-8, 9-10	12 by 11-3/4 by 9-3/4
	10-11, 11-12	12 by 12-3/4 by 10-1/4
	12-13, 13-14	12 by 13-1/4 by 10-1/2
	14-15	12 by 13-3/4 by 10-3/4

[1] Furnished as a guide only.

5.2.1.2 *Type III.* Two hundred and eighty-eight pairs of heels of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container hereinafter specified.

5.2.1.3 *Shipping container.* Each fiberboard shipping container shall conform to style FOL, V3c, or V3s of PPP-B-636. Alternatively, shipping containers shall conform to style OSC, V13c of MIL-B-43291. Each container shall be waterproofed in accordance with the appendix of PPP-B-636. Each shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the applicable container specification.

5.2.1.4 *Mixed sizes.* Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one description only may be packed within the same container.

5.2.2 *Level B.*

5.2.2.1 *Types I and II.* One hundred and forty-four pairs of type I heels or 72 pairs of type II heels of one description only shall be packed in a fiberboard shipping container as hereinafter specified. The approximate inside dimensions of each shipping container shall be as specified in table V. The heels, in pairs, shall be arranged on end within the shipping container as specified in 5.1.1.2, except that type II heels shall be arranged 6 pairs in length.

5.2.2.2 *Type III.* Two hundred and eighty-eight pairs of heels of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container as hereinafter specified.

5.2.2.3 *Shipping container.* Each shipping container shall conform to style OSC (minimum 2 inch overlap), type CF or SF, class domestic, variety SW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II in the appendix of the container specification.

5.2.2.3.1 When specified, shipping containers shall be fabricated from V3c or V3s fiberboard (see 6.2).

5.2.2.4 *Mixed sizes.* Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one description only may be packed within the same container.

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5.2.3 *Level C.* Heels, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2.1 *Labels, mixed sizes.* Each shipping container packed with mixed sizes of one description only shall have securely attached to the end and side a white paper label 5 by 4 inches and with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity of pairs and sizes contained therein.

6. NOTES

6.1 **Intended use.**

6.1.1 Type I heels covered by this specification are intended for use on men's dress shoes.

6.1.2 Type II heels covered by this specification are intended for use on field type of men's footwear.

6.1.3 Classes 3, 4, 5, 6, 7, and 8 will not leave a permanent mark when scuffed over a clean surface.

6.1.4 Classes 2, 4, 5, and 7 will withstand deterioration when in contact with oil.

6.1.5 Class 4 containing cork is slip-resistant and is for use on a safety toe type man's field shoes.

6.1.6 Types I and II, subclass (a) heels are high abrasive commercial heels expected to endure the life of the footwear.

6.1.7 Type III heels covered by this specification are intended for use as components for women's semidress shoes.

6.1.8 All heels covered by this specification are intended for use in repair or manufacture of boots and shoes.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, style, size, and color of heel required (see 1.2).
(*c*) When subclass (a) heels are required (see 1.2.1.1).
(*d*) Whether preproduction sample is required (see 3.1).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(*f*) Whether V-board shipping containers are required for level B shipments (see 5.2.2.3.1).

6.3 Correlation between B & S (USMC) sizes and dealer sizes is as follows:

B & S (USMC)	Dealer	
	Type I	Type II
24	7-8	—
25	—	7-8
26	9-10	—
27	—	9-10
28	10-11	—
29	—	10-11
30	11-11	—
31	—	11-12
32	11-12	—
33	—	12-13
34	12-13	—
35	—	13-14
36	13-14	—
37	—	14-15
38	14-15	—
39	—	15-16
40	15-16	—
41	—	16-17
42	16-17	—

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6.4 The method of packaging referred to in 5.1.1 is similar to method III of MIL-P-116.

6.5 When level B packaging requirements are specified for civil agency procurements, the requirements in 5.1.1 shall apply.

MILITARY CUSTODIANS:
Army—GL
Navy—SA
Air Force—67

Review activities:
Navy—SA, MC
Air Force—67

User activity:
Navy—CG

Preparing activity:
Army—GL

CIVIL AGENCY INTERESTS:
Agr—OPO
Com—NBS
DC—GA
GSA—FSS
HEW—GS
Int—MO
Jus—FPI
VA—DMS

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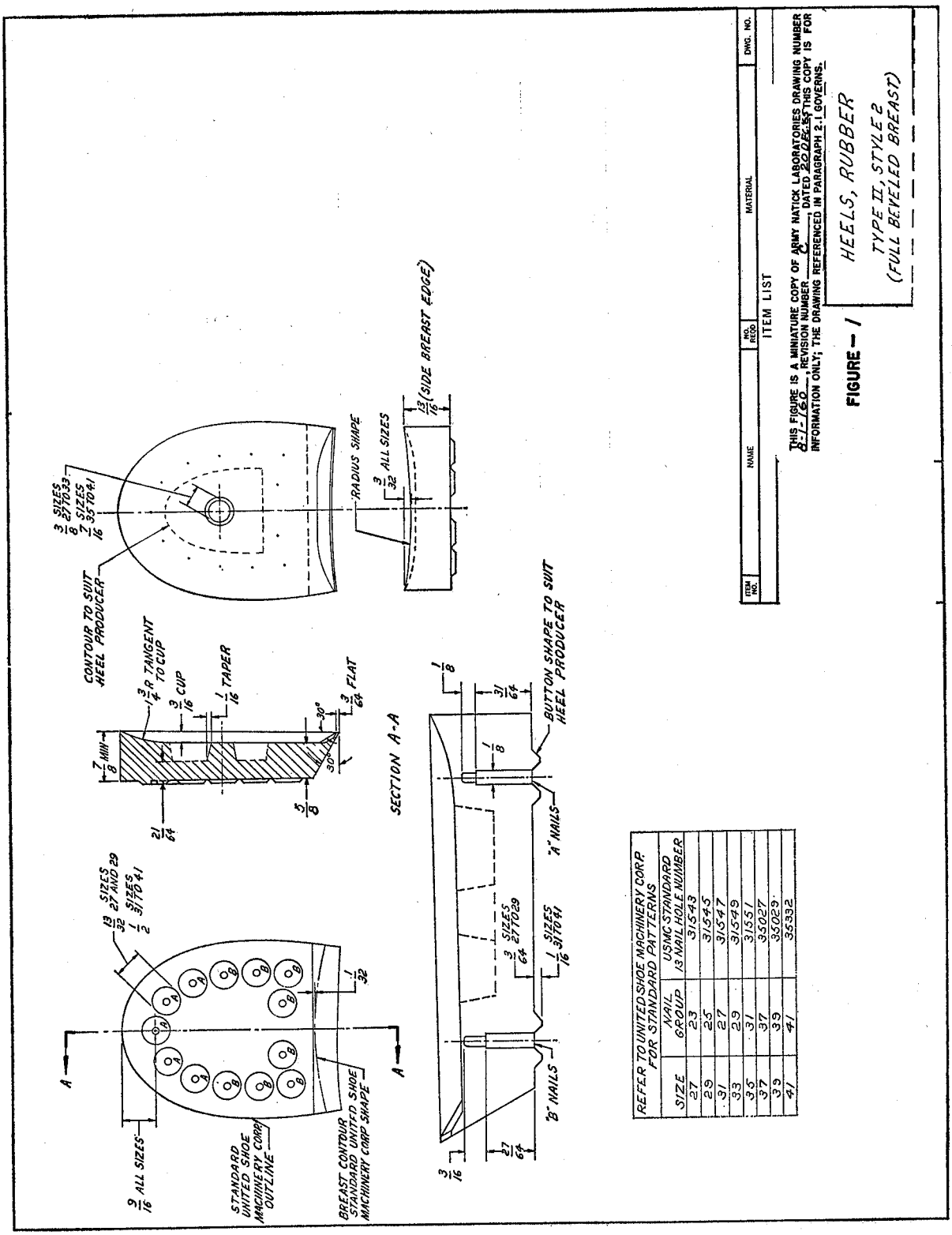
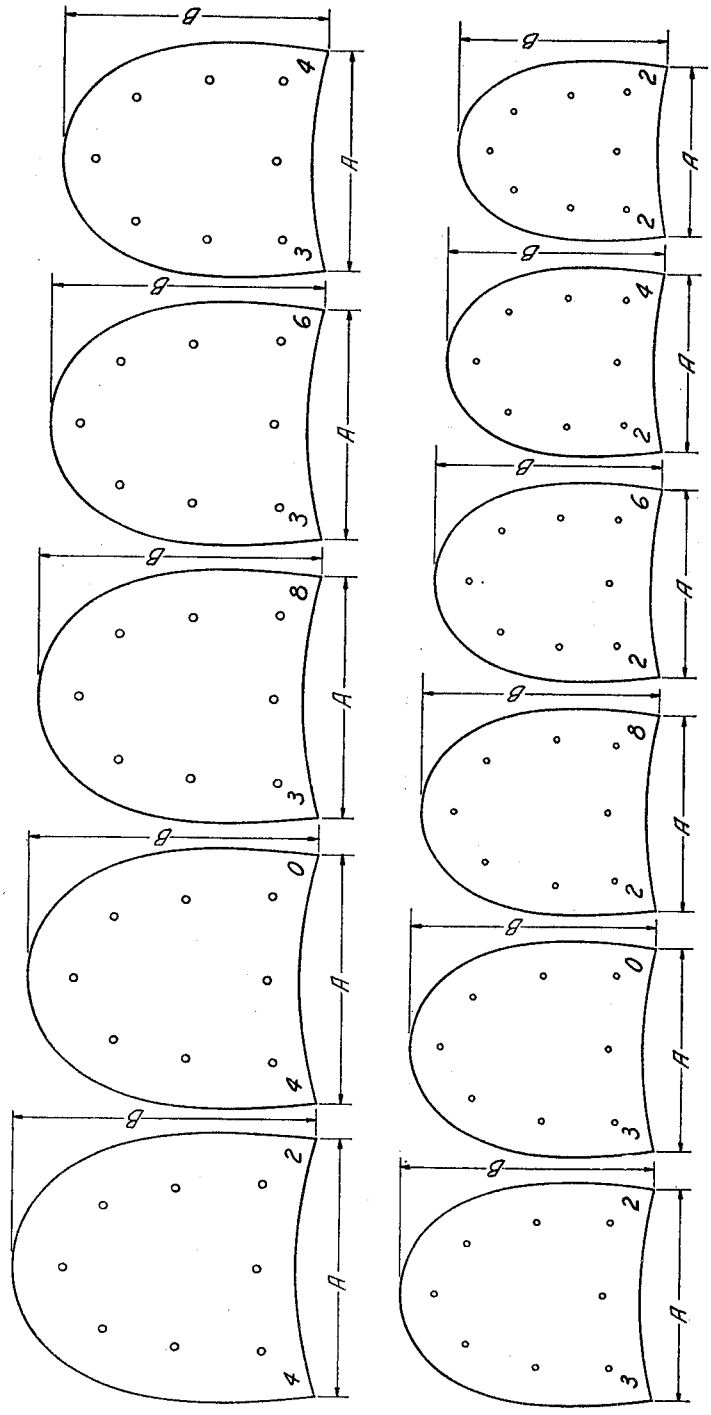


FIGURE – 1

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UNLESS OTHERWISE STATED; FABRICATING TOLERANCES ARE: DECIMALS — WHOLE NUMBERS AND FRACTIONS — 1/64 DEGREES —

FIGURE – 2 STANDARD DIMENSIONS MEN'S HALF HEELS

SIZE	A	B
22	2 3/8"	2 1/8"
24	2 1/2"	3"
26	2 5/8"	3 1/8"
28	2 7/8"	3 1/4"
30	2 7/8"	3 3/8"
32	2 7/8"	3 1/4"
34	3 1/8"	3 5/8"
36	3 3/8"	3 3/4"
38	3 5/8"	3 7/8"
40	3 7/8"	4"
42	3 7/8"	4 1/8"

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Heels, Rubber ZZ-H-141c

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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